**Sanjeetha**
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**Citizen**

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**Professional Summary**

- Over 7+ years' experience in designing, developing, and implementing strategic methods to efficiently solve Big Data processing requirements and Orchestrating data pipelines.
- AI/ML Developed and deployed machine learning models for predictive analytics.
- Proficient in Google Cloud Platform services for data processing and analytics, including BigQuery and Dataflow.
- Expertise in **AWS** services like S3, Redshift, and EMR for scalable data solutions in retail analytics.
- Skilled in using **Azure Data Lake, Azure SQL Database, and Azure Synapse** for financial data analytics.
- Implemented big data solutions using Hadoop and Spark, processing datasets of over 100TB.
- Developed Python scripts for data manipulation and machine learning model training.
- Applied deep learning algorithms for image recognition in the retail industry, enhancing product quality control.
- Utilized **NLP** for sentiment analysis in customer feedback for retail brands.
- Designed and implemented cloud-native**ETL** pipelines using **AWS Glue** and **Azure Data** Factory.
- Managed NoSQL databases like Cassandra and MongoDB, optimizing data storage for large-scale applications.
- Created interactive dashboards using Tableau and **Power BI,** providing actionable insights into finance.
- Leveraged Snowflake for efficient data warehousing solutions in insurance data analytics.
- Used Informatica for robust data integration and transformation processes in banking.
- Skilled in using **DBT** for transforming and modeling data effectively in the food industry analytics.
- Ensured HIPAA compliance in healthcare data processing and analytics.
- Implemented real-time data streaming using Kafka for immediate data insights in retail.
- Built scalable machine learning solutions on **GCP, AWS, and Azure.**
- Conducted time series analysis for financial forecasting and market trend analysis.
- Applied text analytics for enhancing customer service experiences in insurance.
- Developed **AI** models for detecting fraud in e-commerce transactions.
- Created predictive models for patient readmission risks in healthcare.
- Ensured cloud security and compliance across **GCP**, **AWS**, and **Azure** environments.
- Designed recommender systems for personalized customer experiences in retail.
- Implemented data governance frameworks in banking, ensuring data accuracy and integrity.
- Utilized Google BigQuery for large-scale data analytics in various industries.
- Automated data pipelines for efficient data flow and processing in insurance.
- Designed and managed data lakes **in AWS, Azure, and GCP** for centralized data storage.
- Implemented **MLOps** practices for efficient machine learning model lifecycle management.
- Explored blockchain applications for secure transactions in finance.
- Analyzed clinical trial data for pharmaceutical developments in healthcare.
- Conducted customer segmentation analysis to tailor marketing strategies in retail.
- Developed algorithmic trading models in **Python**, enhancing trading strategies in finance.
- Built risk assessment models for accurate insurance premium calculations.
- Analyzed sensor data for optimizing food processing techniques.
- Ensured high data quality standards in all phases of data processing.
- Implemented predictive maintenance models, reducing downtime in manufacturing.
- Developed AI-driven chatbots for enhanced customer engagement in banking.
- Analyzed social media data for brand perception insights in various industries.
- Led cross-functional teams in large-scale technology projects, aligning with corporate goals and visions.
- Championed Agile methodologies, leading to a 30% increase in project delivery speed and efficiency.

- Managed multiple projects simultaneously, consistently delivering within time and budget constraints.
- Built and mentored diverse tech teams, fostering an inclusive and innovative work environment.
- Skilled in conflict resolution, ensuring team cohesion and effective collaboration.
- Successfully engaged with stakeholders at all levels, ensuring alignment and satisfaction with project outcomes.
- Led change management initiatives, introducing new technologies and practices with minimal disruption.
- Proactively managed project risks, reducing potential impacts on timelines and quality.
- Utilized data-driven approaches for strategic decision-making, enhancing project outcomes.
- Maintained a strong focus on customer needs, resulting in higher customer satisfaction rates.
- Advocated for and implemented continuous improvement practices in all projects.
- Demonstrated strong creative problem-solving skills, tackling complex challenges effectively.
- Fostered collaboration between technical and non-technical departments, enhancing project synergy.
- Adapted quickly to changing technologies and market demands, keeping projects relevant and cutting-edge.
- Effectively managed project budgets, optimizing resource allocation and expenditure.
- Applied emotional intelligence in leadership, understanding and managing team dynamics for better
- performance.
- Managed multicultural teams, leveraging diverse perspectives for enriched problem-solving.
- Implemented effective feedback mechanisms, enabling continuous learning and development.
- Optimized resource utilization across projects, maximizing team productivity and project outcomes.
- Excelled in strategic planning and execution, ensuring long-term project success and alignment with business objectives.
- Demonstrated excellent time management and prioritization skills, handling high workloads efficiently.
- Maintained a strong focus on quality assurance, delivering projects that meet and exceed standards.
- Initiated and supported professional development programs for team skill enhancement.
- Practiced transparent communication, building trust and credibility with teams and stakeholders.
- Successfully managed remote teams, ensuring productivity and engagement in a virtual environment.
- Effective in crisis management, quickly resolving unforeseen issues with minimal impact on project continuity.

**Technical Skills**

- Cloud Platforms: AWS (S3, Redshift, EMR, Glue, Lambda), GCP (BigQuery, DataProc, DataFlow, Composer), Azure (Data Lake, Synapse, ADF)
- Big Data Tools: Hadoop, Spark (PySpark, Scala), Kafka, Hive, HBase
- Programming & Scripting: Python, Scala, Shell Scripting (Bash), PL/SQL
- Data Integration & Warehousing: Snowflake, DBT, Informatica, PostgreSQL, MongoDB, Cassandra
- Visualization: Tableau, Power BI, AWS QuickSight
- DevOps &IaC: Terraform, Jenkins, CloudFormation
- Methodologies: Agile (Scrum), MLOps, Data Governance Frameworks

**Education**

- **Bachelor of Arts from Kakatiya University**

**Work Experience**

**March 2022 to Till Date**
**Sun Life Financial, Wellesley Hills, MA**
**Senior Data Engineer**

- Designed and developed several data pipelines to streamline data ingestion and transformation processes in Azure. I worked on end-to-end ETL and ELT workflows, utilizing tools like Azure Data Factory, Azure Synapse Analytics, and Azure Databricks.
- Developed Terraform script and deployed it in the cloud deployment manager to spin up resources like cloud virtual networks.
- Used Spark-SQL to load JSON data and create Schema RDD and loaded it into Hive Tables and handled structured data using Spark SQL.
- Computed Engines in public and private subnets along with AutoScaler in Google Cloud Platform.
- Created firewall rules to access Google DataProc from other machines.
- Used Python to ingest data into BigQuery.
- Involved in porting the existing on-premise Hive code migration to GCP (Google Cloud Platform) BigQuery.
- Fostered collaboration between development, operations, and healthcare client stakeholders, resulting in a 25% improvement in project delivery efficiency.
- Successfully navigated healthcare compliance audits, achieving a 95% compliance score, and ensuring adherence to industry regulations.
- Integrated AWS Snowflake to create a robust and scalable data warehouse, enabling advanced analytics for customer behavior and inventory optimization in retail.
- Spearheaded innovative approaches to healthcare DevOps, ensuring the integration of the latest technologies and methodologies for continual improvement in healthcare systems.
- Pioneered the integration of data engineering and DevOps practices, fostering a collaborative environment that resulted in a 25% increase in overall project efficiency.
- Automated end-to-end data pipelines, reducing deployment times by 30% and ensuring seamless data flow for healthcare applications.
- Azure Cloud Services Management: Extensively utilized Microsoft Azure services for deploying and optimizing finance applications, enhancing system performance and scalability.
- AI-Powered Demand Forecasting: Developed AI models for accurate demand forecasting in retail, significantly improving inventory management.
- Machine Learning for Customer Segmentation: Implemented machine learning algorithms for customer segmentation, aiding in personalized marketing strategies.
- Data Warehousing Design and Management: Designed and managed robust data warehousing solutions, supporting comprehensive finance and insurance data analysis.
- Efficient Data Movement Strategies: Orchestrated efficient data movement processes, ensuring timely and accurate data availability for finance decision-making.
- Data Transformation for finance Insights: Performed complex data transformations, enabling deeper insights into finance and customer behavior.
- Automated Data Ingestion Processes: Implemented automated data ingestion pipelines, enhancing the speed and reliability of data flow into finance and insurance systems.
- Data Modeling for finance Analytics: Developed sophisticated data models tailored to insurance business needs, facilitating effective data-driven strategies.
- Data Encryption and Security Protocols: Ensured high levels of data security in finance environments through rigorous data encryption and security protocols.
- Implemented scalable and cost-effective data storage solutions using Azure Data Lake Storage, ensuring seamless access and analysis of large datasets.
- Real-time Data Streaming with GCP: Managed real-time data streaming services on GCP, enabling immediate analysis and response to finance and insurance trends.
- NLP for Customer Feedback Analysis: Applied natural language processing (NLP) techniques to analyze customer

feedback, enhancing customer satisfaction and loyalty.

- Finance Data Visualization with BI Tools: Created dynamic and interactive finance dashboards using business intelligence tools, providing actionable insights to management.
- Cloud-Based ETL Pipeline Optimization: Optimized ETL pipelines in the cloud, improving data processing efficiency and accuracy.
- Monitored and optimized the performance of data pipelines using Azure Monitor and Application Insights, enabling proactive identification and resolution of bottlenecks.
- Finance-Specific AI Chatbots: Developed AI-powered chatbots for enhancing customer service and engagement.
- in finance environments.
- Customer Behavior Analysis with ML: Analyzed customer behavior using machine learning models, enabling targeted marketing and sales strategies.
- Inventory Optimization Algorithms: Created algorithms for inventory optimization, reducing overheads and improving stock turnover.
- SQL and NoSQL Database Management: Expertly managed both SQL and NoSQL databases, catering to diverse data needs in the finance and insurance sector.
- Orchestrated security measures for healthcare data, achieving a 20% reduction in vulnerabilities and ensuring compliance with healthcare data regulations.
- Enabled continuous integration of data assets, reducing data integration time by 25% and providing real-time insights for healthcare analytics.
- Instituted robust data quality checks, resulting in a 15% improvement in data accuracy and reliability for healthcare analytics.
- Applied DevOps principles to big data projects, achieving a 30% reduction in time-to-market for healthcare data-driven applications.
- Implemented real-time data streaming solutions, providing instant insights for healthcare monitoring systems and reducing latency by 25%.
- Automated infrastructure deployment using AWS CloudFormation, ensuring consistent and repeatable provisioning of cloud resources for big data and DevOps projects.
- Established a data governance framework, ensuring data integrity and facilitating collaboration between data engineers and DevOps teams for healthcare projects.
- Implemented Infrastructure as Data Code (IaDC) principles, treating data infrastructure as code to enhance reproducibility and scalability for healthcare data platforms.
- Implemented advanced data monitoring and alerting systems, reducing downtime by 20% and ensuring the reliability of healthcare data pipelines.
- Automated data ETL processes, resulting in a 25% reduction in manual intervention and ensuring timely delivery of accurate data for healthcare analytics.

**Environment:** Google Cloud Platform (GCP), Amazon S3, AWS, Azure, Azure Data Lake Storage, Spark, Hive, Hadoop, Google DataProc, Python, Scala, Shell Script, BigQuery, SQL, NoSQL, Terraform, Git, Jenkins, Terraform, Git, Jenkins, IaC, IaDC.

**November2019 to February 2022**
**Universal Health Services Inc,Kings of Prussia, PA**
**Data Engineer**

- Built and architected multiple data pipelines, implementing end-to-end ETL and ELT processes for data ingestion and transformation using AWS services such as AWS Glue, Amazon S3, and AWS Lambda, while coordinating tasks among the team for seamless project execution.
- Loaded Data to BigQuery using Google DataProc, GCS bucket, HIVE, Spark, Scala, Python, Gsutil and Shell Script.
- Developed Terraform script and deployed it in cloud deployment manager to spin up resources like cloud virtual networks.
- Used Spark-SQL to load JSON data and create Schema RDD and loaded it into Hive Tables and handled

structured data using Spark SQL.

- Computed Engines in public and private subnets along with AutoScaler in Google Cloud Platform.
- Created firewall rules to access Google DataProc from other machines.
- Used Python to ingest data into BigQuery.
- Involved in porting the existing on-premises Hive code migration to GCP (Google Cloud Platform) BigQuery.
- Involved in extracting, transforming, and loading data sets from local to HDFS using Hive.
- Imported and exported data from RDBMS to HDFS and vice-versa using Sqoop.
- Used Azure Databricks to load and transform JSON data, create Schema RDD, and store results in Azure SQL Database and Azure Blob Storage for downstream analytics.
- Involved in writing Pig scripts to analyze or query structured, semi-structured and unstructured data in a file.
- Written Python programs to maintain raw file archival in the GCS bucket.
- Written Scala programs for Spark transformation in DataProc.
- Created complex schema and tables for analyzing using Hive.
- Set required Hadoop environments for cluster to perform Map Reduce jobs.
- Formatted data using Hive queries and stored on HDFS.
- Built a program with Python and Apache Beam and executed it in cloud Dataflow to run Data validation between raw source file and BigQuery tables.
- Automated deployment and management of AWS infrastructure using Terraform and AWS CloudFormation templates, ensuring consistent and repeatable environments.
- Built a Scala and Spark based configurable framework to connect common Data sources like MySQL, Oracle, PostgreSQL Server, Salesforce and load it in BigQuery.
- Monitored BigQuery, DataProc and cloud Dataflow jobs via Stack driver for all the environments.
- Opened SSH tunnel to Google DataProc to access the Yarn manager to monitor Spark jobs.
- Submitted Spark jobs using Gsutil and Spark submission gets it executed in the DataPro cluster.
- Built some complex data pipelines using PL/SQL scripts, Cloud REST APIs, Python scripts, GCP Composer and GCP Dataflow.
- Built ETL systems using Python and in-memory computing framework (Apache Spark), scheduling and maintaining data pipelines at regular intervals in Apache Airflow.
- Developed PySpark and Scala-based transformations on Azure Databricks for high-performance data processing and transformation.
- Worked on HBase and MySQL for optimizing the data.
- Worked over File Sequence, AVRO, and Parquet file formats.
- Monitored Hadoop cluster connectivity and security using tools such as Zookeeper and Hive.
- Manage and review Hadoop log files.
- AGILE(Scrum) development methodology has been followed to develop the application.
- Involved in managing the backup and disaster recovery for Hadoop data.
- Configured SSH access to Azure Virtual Machines for secure management and troubleshooting of Spark and Hadoop workloads.
- Performed data wrangling to clean, transform and reshape the data utilizing panda’s library.
- Analyzed data using SQL, Scala, Python, Apache Spark and presented analytical reports to management and technical teams.
- Data Visualization with Quick Sight: Created interactive dashboards using AWS Quick Sight for real-time visualization of patient data, improving clinical decision-making processes.
- EHR Data Integration: Integrated Electronic Health Records (EHR) with AWS cloud services, streamlining patient data accessibility for healthcare professionals.
- Custom AI Solutions: Developed custom AI algorithms for patient diagnosis and treatment recommendations, improving patient care quality.
- Natural Language Processing: Applied NLP techniques to analyze and interpret clinical notes, enhancing patient data analysis and insights.
- Automated Alert Systems: Implemented automated alert systems using AWS SNS for critical patient monitoring, reducing response times in emergency situations.
- Serverless Architecture: Utilized AWS Lambda for building serverless applications, reducing operational costs

and improving scalability.

**Environment:** GCP, AWS, Azure, Hadoop, Hive, Spark, Apache Beam, DataProc, DataFlow, HBase, Python, Scala, PL/SQL, Shell Scripting, BigQuery, MySQL, PostgreSQL, Oracle, Salesforce, Apache Airflow, AVRO, Terraform, WS Lambda

**February 2017 to September 2019**
**Costco, Issaquah, WA**
**Data Engineer**

- Involved in the requirement gathering phase to align business user needs with data processing solutions, ensuring flexibility for evolving project requirements in AWS.
- Responsible for managing data from multiple sources.
- Designed a Data Quality Framework to perform schema validation and data profiling on Spark (PySpark).
- Developed ETL pipelines using Azure Data Factory, transforming data using Spark and Python, and storing the results in Azure Data Lake or SQL Database.
- Developed Spark code using Scala and Spark-SQL/Streaming for faster testing and processing of data.
- Implemented usage of Amazon EMR for processing Big Data across a Hadoop Cluster of virtual servers on
- Amazon Elastic Compute Cloud (EC2) and Amazon Simple Storage Service (S3).
- Deployed the project on Amazon EMR with S3 connectivity for setting a backup storage.
- Well-versed in using Elastic Load Balancer for Autoscaling in EC2 servers.
- Designed data and ETL pipeline using Python and Scala with Spark.
- Developed highly complex Python and Scala code, which is maintainable, easy to use, and satisfies application requirements, data processing and analytics using inbuilt libraries.
- Involved in designing optimizing Spark SQL queries, Data frames, import data from Data sources, performing transformations, perform read/write operations, save the results to output directory into HDFS/AWS S3.
- Migrated MapReduce programs into Spark transformations using PySpark.
- Developed and optimized data extraction, transformation, and aggregation using Azure Data Factory pipelines, supporting various data formats like JSON, Parquet, and CSV.
- Configured Spark streaming to get ongoing information from Kafka and stored the stream information to HDFS.
- Experience with Spark Context, Spark-SQL, Spark, YARN.
- Designed and implemented a Data Quality Framework leveraging AWS Glue and AWS Lambda for schema validation and data profiling of incoming data.
- Implemented Spark-SQL with various data sources like JSON, Parquet, ORC and Hive.
- Implemented PySpark Scripts, Spark-SQL to access Hive tables into Spark for faster processing of data.
- Developed Pig Latin scripts to extract the data from the web server output files to load into HDFS.
- Built dashboards using Tableau to allow internal and external teams to visualize and extract insights from big data platforms.
- Responsible for analyzing and cleansing raw data by performing Hive queries and running Pig scripts on data.
- Created Hive tables, loaded data and wrote Hive queries that run within the map.
- Worked on ETL Processing which consists of data transformation, data sourcing and mapping, Conversion, and
- loading.
- API Integration for Retail Systems: Integrated various APIs to enhance functionality and interoperability of retail systems and applications.
- Optimized MapReduce Jobs to use HDFS efficiently by using various compression mechanisms.
- Used Amazon Elastic Cloud Compute (EC2) infrastructure for computational tasks and Simple Storage Service (S3) as storage mechanism.
- Worked with building data warehouse structures, and creating facts, dimensions, aggregate tables, by dimensional modeling, Star and Snowflake schemas.
- Worked on Dockers containers by combining them with the workflow to make them lightweight.
- Written multiple MapReduce programs for data extraction, transformation and aggregation from multiple file formats including XML, JSON, CSV and other compressed file formats.

- Worked with AWS Redshift and DynamoDB for handling structured and unstructured data, leveraging SQL and NoSQL technologies.
- Worked on Kafka REST API to collect and load the data on Hadoop file system and used Sqoop to load the data from relational databases.
- Worked on MongoDB by using CRUD (Create, Read, Update and Delete), Indexing, Replication and Shading features.
- Predictive Analytics for Sales Trends: Utilized predictive analytics to identify and capitalize on emerging sales trends, boosting retail revenue.
- Worked on SQL queries in dimensional data warehouses and relational data warehouses. Performed Data Analysis and Data Profiling using Complex SQL queries on various systems.
- Big Data Analytics for Retail Insights: Leveraged big data analytics to extract meaningful insights from large volumes of retail data, driving strategic decisions.
- Followed agile methodology for the entire project. Data Governance in Retail Settings: Established and maintained robust data governance frameworks, ensuring data integrity and compliance in retail operations.

**Environment:** AWS, Azure, Hadoop, Hive, Spark, Pig, Kafka, MongoDB, Python, Scala, SQL, Shell Scripting, Spark Streaming, YARN, MapReduce, JSON, Parquet, ORC, CSV, XML, Tableau, Sqoop, Apache Spark, PySpark, Kafka REST API, Docker, Big Data Analytics, Data Governance Frameworks